an overeager microwave oven in the lunch room.) Penzias and Wilson examined every inch of the detector, and even considered the possibility that a small group of pigeons nesting in the antenna could be the source of the hum.[fn8] But no matter what they did, they couldn't get the hum to go away, and they never found any interference that could account for it. So they had to consider the possibility that it really was coming from space, and from every direction in the sky. But what could it be? Anything coming from the planets or the Sun should only show up at certain times and directions, and even emission from our own Milky Way galaxy would not be completely uniform.

Enter the Princeton team. In a roundabout way.

To backtrack a moment, Peebles's calculations had said that if the universe was hot everywhere early on, then we should be currently awash in the leftover radiation from it. Here's what he was thinking. If looking farther away means looking farther into the past, and if there was a time in the distant past when the universe was basically one big all-encompassing fireball, then it should be possible to look so far away that you see a part of the universe that is *still on fire*. Or, thinking about it another way: if, 13.8 billion years ago, the whole possibly infinite universe was aglow with radiation, there should be parts of it so far away that the radiation from that glow is only just now reaching us, having traveled through the expanding, cooling space all this time. In any direction we look, if we look far enough away, we'll see that distant fiery universe. We're not looking at parts of *space* that are any different, but rather at a *time* when ALL of space was on fire.

So, this background radiation should come from everywhere. And it should come from everywhere no matter where you are, because you can always look far enough away to see the hot phase of the cosmos. The speed-of-light/time-travel connection gives you this for free. Every point in space is the center of its own sphere of ever deepening time, bounded by a shell of fire.

Peebles realized this, and, as physicists tend to do, talked to his colleagues about his extremely mind-blowing thoughts. He even passed around a preprint of a paper in which he described what he and his colleagues planned to do to detect this radiation. Then, word proceeded to travel the 60 kilometers to Bell Labs—via two unconnected physicists, an airplane, and Puerto Rico.